Digital Learning Platform for Rural School Students in Nabha SIH25019

Abstract

Education plays a crucial role in social and economic development, yet many rural schools face challenges in accessing quality educational resources and modern learning tools. Limited digital infrastructure, lack of structured online platforms, and restricted access to learning materials often affect the quality of education available to rural students. To address these challenges, a Digital Learning Platform for Rural School Students is proposed to provide an organized and accessible web-based learning environment.

The proposed system is designed as a centralized digital platform where students, instructors, and administrators can interact through dedicated dashboards. Instructors can upload study materials, assignments, and course content, while students can access learning resources and submit assignments through the platform. The system also includes analytics dashboards that help monitor learning activities and track student engagement.

The platform is developed using modern web technologies such as Python, Django framework, HTML, CSS, and JavaScript, ensuring a secure and scalable system architecture. The system uses SQLite database to manage user information, course materials, and learning activity records. In addition, a Bidirectional Long Short-Term Memory (BLSTM) model is integrated to perform sentiment analysis on user feedback, helping evaluate the learning experience and improve platform usability.

By combining web technologies with intelligent data analysis, the proposed platform aims to enhance digital learning accessibility for rural students, support effective course management for instructors, and improve the overall educational experience through a structured and interactive learning environment.

Introduction

The rapid advancement of digital technologies has transformed the way education is delivered and accessed. Online learning platforms enable students to access study materials, assignments, and educational resources anytime through web-based systems. However, many rural schools still rely on traditional teaching methods and lack structured digital learning platforms. The proposed Digital Learning Platform for Rural School Students in Nabha is designed to provide a centralized web-based system where instructors can upload course materials and students can access learning resources through interactive dashboards. The platform uses technologies such as Python, Django, HTML, CSS, and JavaScript, along with a SQLite database, to create a secure and efficient digital learning environment.

Motivation

The project is driven by:

- Provide digital learning resources for rural students
- Help students access study materials easily online
- Improve interaction between instructors and students
- Support modern learning using a web-based platform

Scope

The proposed system provides a web-based platform for managing and sharing educational content. Administrators manage users, instructors upload course materials, and students access learning resources through their dashboards. The system also includes analytics and **BLSTM-based sentiment analysis** to evaluate user feedback and improve the learning experience.

Problem Statement

1.2 Problem Identification

Existing educational systems in many rural schools face several limitations that affect the accessibility and quality of learning resources:

1. **Limited Access to Digital Learning Resources** – Many rural schools lack structured digital platforms where students can access study materials, assignments, and educational content outside classroom hours.
2. **Dependence on Traditional Teaching Methods** – Learning mainly relies on classroom lectures and printed materials, limiting exposure to interactive and modern learning tools
3. **Difficulty in Sharing Educational Content** – Instructors often face challenges in distributing study materials, assignments, and updates efficiently to all students.
4. **Lack of Centralized Learning Platform** – Educational content and student activities are not managed through a unified system, making it difficult to organize and monitor learning resources
5. **Limited Communication Between Students and Instructors** – Outside classroom environments, students have fewer opportunities to interact with instructors or clarify doubts.

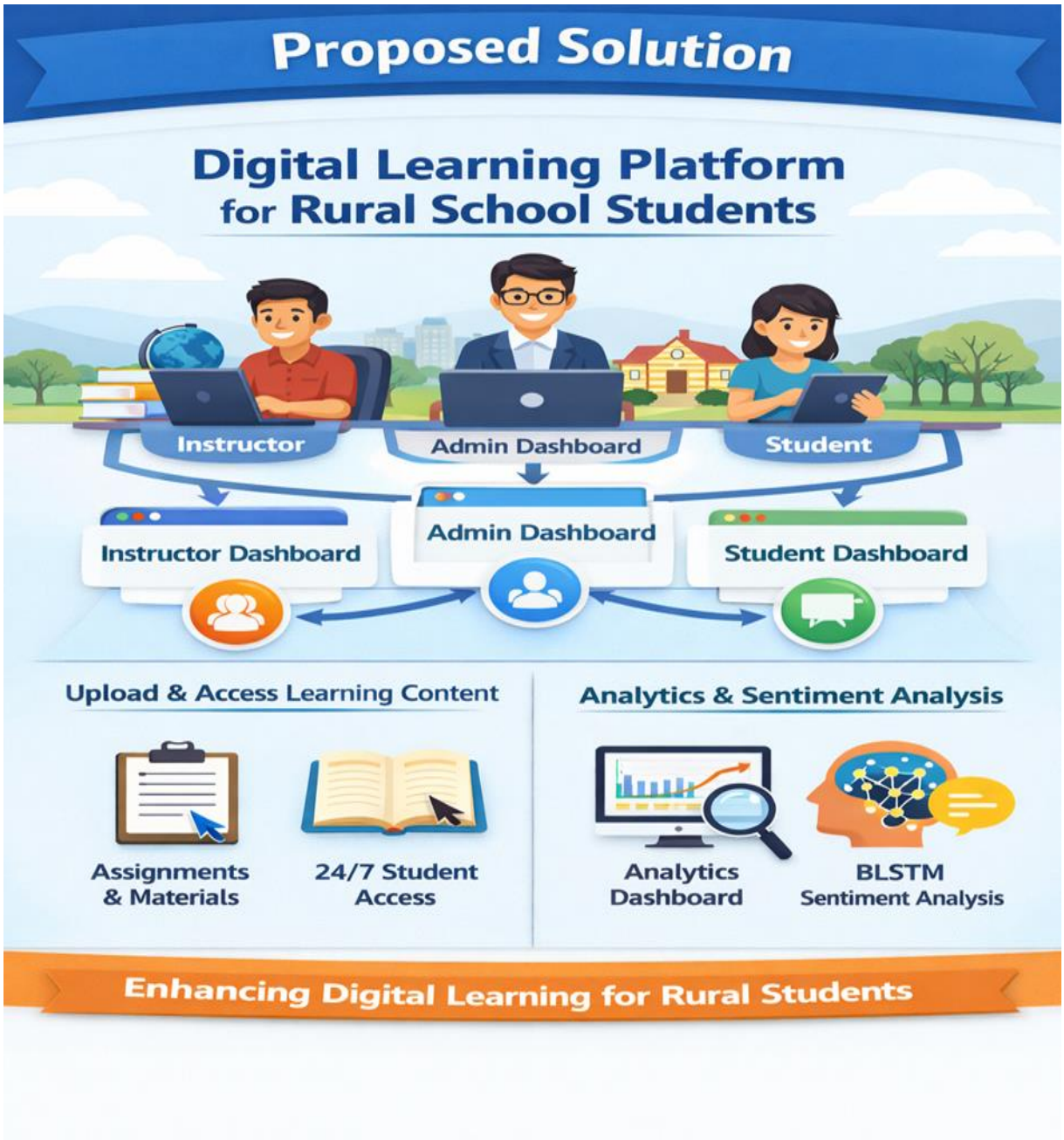
6. **Low Adoption of Modern Educational Technologies** – Many rural educational institutions have limited access to modern digital tools that support flexible and interactive learning..

These challenges reduce the effectiveness of learning and create a gap between rural and urban education systems. Therefore, a centralized **Digital Learning Platform** is required to provide structured access to learning resources and improve educational accessibility for rural students.

Proposed Solution

To address the challenges faced by rural schools in accessing digital learning resources, a **Digital Learning Platform for Rural School Students** is proposed. The system is designed as a web-based application that provides separate dashboards for administrators, instructors, and students. Instructors can upload study materials, assignments, and course content, while students can access learning resources and educational materials through the platform anytime. This helps create a centralized environment for managing and sharing educational content efficiently.

The platform also includes features such as analytics dashboards to monitor learning activities and student engagement. In addition, **BLSTM-based sentiment analysis** is used to analyze user feedback and understand the learning experience of students. By integrating modern web technologies with intelligent data analysis, the proposed system aims to improve accessibility to educational resources and enhance digital learning


Proposed Solution
Digital Learning Platform
for Rural School Students
Instructor
Admin Dashboard
Student
Instructor Dashboard
Admin Dashboard
Student Dashboard
Upload & Access Learning Content
Assignments
& Materials
24/7 Student
Access
Analytics & Sentiment Analysis
Analytics
Dashboard
BLSTM
Sentiment Analysis
Enhancing Digital Learning for Rural Students

The proposed system provides a structured digital learning platform that enables instructors to upload course materials, assignments, and educational resources while allowing students to access learning content through a centralized web interface. By organizing educational resources in a single platform, the system helps students easily access study materials and supports flexible learning beyond traditional classroom environments.

Another important component of the system is the **dashboard module**, which provides separate dashboards for administrators, instructors, and students. These dashboards help manage learning activities efficiently by displaying information such as uploaded materials, assignments, and user activity. Visual analytics tools also help monitor learning engagement and track overall platform usage.

The system also includes **secure authentication and role-based access control** to ensure that only authorized users can access specific features of the platform. Administrators can manage users, instructors can upload and manage learning materials, and students can access course content through their dashboards. These features help maintain system security and ensure organized management of educational data.

In addition, the system incorporates **BLSTM-based sentiment analysis** to analyse user feedback and understand the learning experience of students and instructors. By evaluating feedback patterns, the platform can help identify areas for improvement and enhance the overall learning environment.
Overall, the proposed system aims to create an efficient and user-friendly digital learning platform that improves access to educational resources, supports effective course management, and enhances digital learning opportunities for rural school students.

KEY FEATURES

1. **Role-Based Dashboards** – Provides separate dashboards for **Admin, Instructor, and Student** to manage activities efficiently.

2. **Learning Material Management** – Instructors can upload and manage **course materials, notes, and assignments** on the platform.

3. **Easy Access to Resources** – Students can access **study materials, assignments, and course content anytime** through the web platform.

4. **Interactive Analytics Dashboard** – Displays learning activity and platform usage through **visual charts and statistics**.

5. **Secure Authentication System** – Ensures **safe login and role-based access control** for different users.

6. **Sentiment Analysis using BLSTM** – Analyses **user feedback** to understand student experience and improve the learning platform.

SYSTEM ARCHITECTURE

The system architecture of the Digital Learning Platform for Rural School Students represents the overall structure of the platform and the interaction between its different modules. The architecture is designed to ensure efficient management of educational resources, secure user access, and smooth communication between instructors and students. The system follows a modular design where different components work together to provide a complete digital learning environment.
The architecture begins with the User Interface Layer, which allows users to interact with the system through a web portal. Students, instructors, and administrators can access the platform using a web browser and interact with different features such as viewing course materials, uploading assignments, and managing learning activities. The interface is developed using HTML, CSS, and JavaScript, providing a user-friendly and responsive experience.

The next component is the Application Layer, which handles the main system functionality and processes user requests. This layer is implemented using Python and the Django framework, which manages tasks such as user authentication, course management, assignment handling, and communication between different modules of the platform.
Another important component is the Database Layer, where all system data is stored securely. The database stores user information, course materials, assignments, and learning activity records. The platform uses SQLite database to manage and organize data efficiently
In addition, the system includes a BLSTM-based sentiment analysis module that analyses feedback from users to understand their learning experience and improve the platform. These components work together to create an efficient and secure digital learning platform that enhances accessibility to educational resources for rural students.

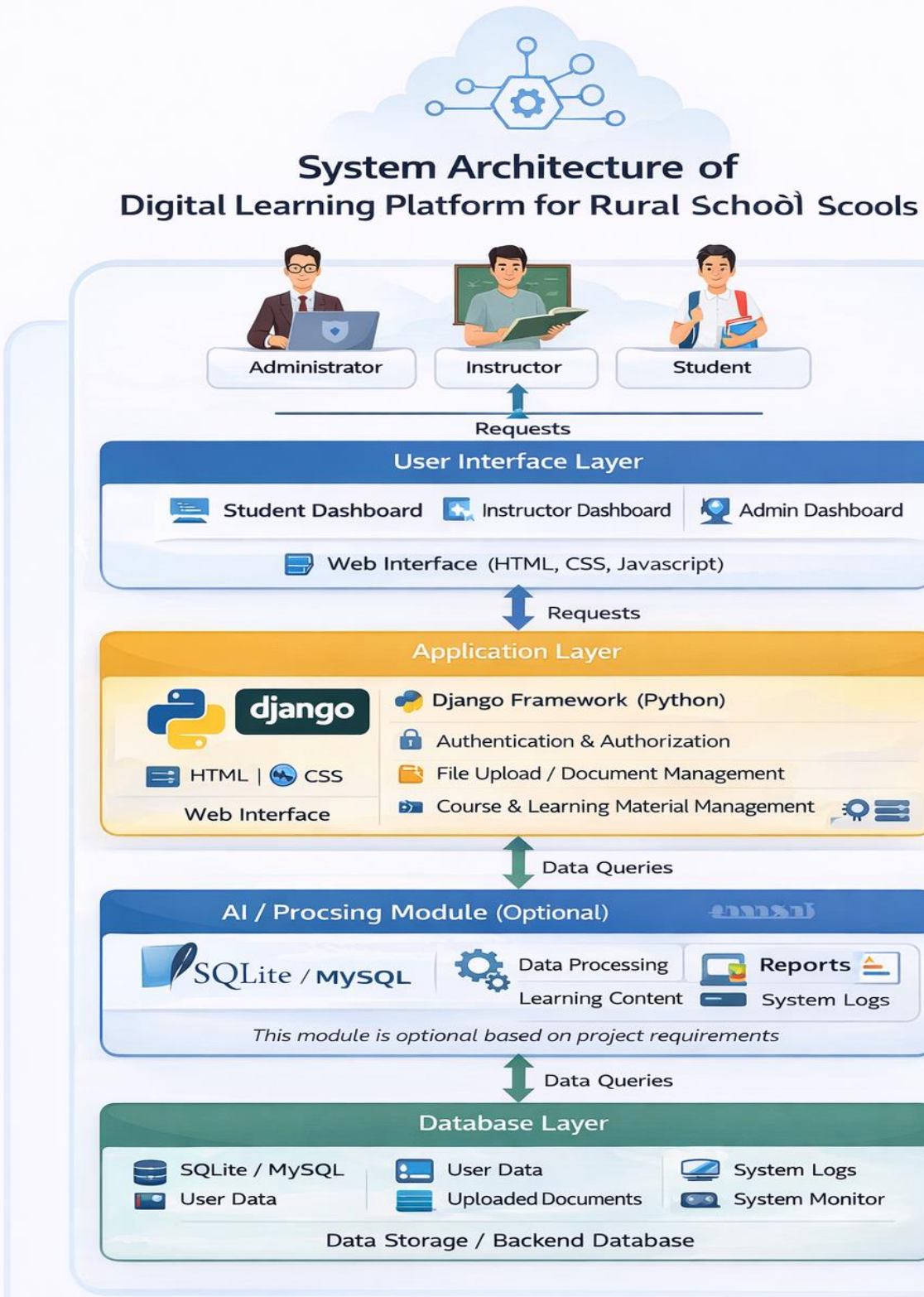


Figure: System Architecture of Digital Learning Platform for Rural Schools

TECHNOLOGIES USED

The Digital Learning Platform for Rural School Students is developed using modern web technologies that ensure efficiency, security, and scalability of the system. The technologies used in this project are described below.

Python

Python is used as the main programming language for backend development. It handles system logic, data processing, and communication between different components of the application.

Django Framework

Django is used as the web framework to manage server-side operations such as request handling, routing, authentication, and application logic. It also provides built-in security features and efficient database management.

HTML, CSS, and JavaScript

These technologies are used to design the user interface and structure of the web pages. They help create a responsive and interactive environment for users to access the platform.

Chart.js / Visualization Tools

Visualization libraries are used to display analytics dashboards, charts, and graphical representations of learning activity and platform usage.

SQLite Database

SQLite is used as the database to store system data such as user information, course materials, assignments, and activity records.

BLSTM (Bidirectional Long Short-Term Memory)

BLSTM is used for sentiment analysis to analyse user feedback and understand student learning experiences. This helps improve the effectiveness of the learning platform.

Impact of the System

The **Digital Learning Platform for Rural School Students** provides an effective solution for improving access to educational resources in rural areas. By offering a centralized platform for managing and accessing learning materials, the system helps students gain better access to study resources beyond traditional classroom environments. This supports continuous learning and enhances the overall educational experience for rural students.

The platform also improves the ability of instructors to share course materials, assignments, and updates efficiently through the web-based system. The use of dashboards and analytics tools helps monitor learning activities and track student engagement, enabling instructors and administrators to understand learning patterns more effectively.

Another important impact of the system is the improvement of communication and interaction between instructors and students. By providing a digital platform where learning materials and assignments can be accessed anytime, the system encourages flexible learning and supports the adoption of modern educational technologies in rural schools.

Overall, the platform contributes to enhancing digital learning opportunities, reducing the educational gap between rural and urban areas, and promoting the use of technology in education.

Conclusion

The Digital Learning Platform for Rural School Students provides a structured approach to support digital education in rural areas. The system enables students to access learning materials and assignments through a web-based platform while allowing instructors to manage educational content efficiently. By organizing educational resources in a centralized system, the platform helps improve accessibility and learning opportunities for students who may have limited access to digital tools.

The implementation of this platform demonstrates how modern web technologies can be used to support education in underserved regions. By improving resource availability, supporting flexible learning, and encouraging the use of digital tools in education, the system contributes to strengthening the overall learning environment for rural students.